

CI Board — Wednesday 2026-07-01 wake

Tuesday EOD called by Casey after long-pull session that broke the “flagged = permanently held” stall.

Honest state going in

Count: 9 baseline (held per Casey hold pending Cal α tier-review). $\theta_{13} = 1/(N_c^2 \cdot n_C) = 1/45$ banked as 10th candidate (team consensus $9 \rightarrow 10$ firm; awaiting Casey formal acknowledgment + Cal α review).

Open dimensionless pieces: 2 (per-channel α + lepton-prefactor; F438 AB-10 unit-definition dissolved mass-unit seal).

4 pure scales at PRINCIPLED FLOOR per Casey #9 (Higgs vev, Λ , G, neutrino scale) — scale-invariance of operator algebra A.

Sin $\theta_C + \theta_{12}$ honestly held as candidates (NOT bank): Lyra’s F442 honest negative — $d(\nu)$ formal degree does NOT predict $m_s/m_d = 20$; the form-match was real, the mechanism isn’t there. Linked pair pending mass-formula derivation.

Substantive Tuesday landings (carry-forward)

- θ_{13} bank candidate (Lyra; $9 \rightarrow 10$): $1/45$ mechanism-disambiguated via boundary-reaching
- F437 (Grace): CKM as ONE geometric object — zero parameters, $\sim 5\text{-}8\%$ structural, $V_{us}=(3/4)^5$, $V_{cb}=\text{dual-}\rho$, $V_{ub}=(1/3)^5$
- F438 (Lyra): AB-10 = unit definition via Casey #9 + Kähler-Einstein $\rightarrow$ mass-unit seal dissolved, open pieces $3 \rightarrow 2$
- F441 (Lyra): α value-banked via Wyler; EM-identification gated on per-channel α deep gate; Elie’s blind-test catch resolved canonically
- F435/F436: Wolfenstein hierarchy operator-level via multiplicative kernel composition
- Cross-sector unification: Cabibbo + PMNS θ_{12} both 1-2 mixings on $\text{dim}_R = 10$, ratio = C_2 exactly
- Elie down-quark ladder (toy 4521): $m_d:m_s:m_b = 1:20:900 = 1:\text{rank}^2 \cdot n_C:(N_c^2 \cdot n_C) \cdot (\text{rank}^2 \cdot n_C)$ — target-innocent candidate
- Sector-specificity capstone (toy 4522): down=integer ladder; leptons= π -forms; up-quarks=g-ladder; no universal mass law — sector-specificity IS the finding
- Cross-link noted: $\sin^2 \theta_{13} = 1/45 = m_s/m_b = 1/(N_c^2 \cdot n_C)$ — mixing and mass-ratio share substrate structure
- Grace K-address geometric anchor: down 1-2 mode-count = $20 = \text{rank}^2 \cdot n_C = S^4$ scalar harmonics $\leq \deg 2$

Team’s consolidated top lanes (Grace + Lyra + Keeper recommendations)

Strongest convergence on down-quark mass formula via Vol-16 + QCD running as Wednesday’s PRIMARY lane:
- Grace A1: reframe $d(\nu)=64 \rightarrow 45$ observed via QCD running $\times 0.70$ (principled mechanism, not fudge) - Lyra #1: $(\text{sector}^2) \cdot n_C$ ladder pattern — WHY rank^2 for 1-2 and N_c^2 for 2-3? - Complementary approaches; same target

Honest yield if lands: down mass ratios (+2 or +3) + sin θ_C via GST from substrate (+1) + θ_{12} linked pair (+1) $\rightarrow$ potential $10 \rightarrow 14$ or higher

Per-CI lanes Wednesday wake

LYRA — Primary: down-quark mass formula derivation

Lane L1 (primary, work until finished): - Vol-16 mass formula via $d(\nu) \times$ measurement determinant - Address Lyra’s own $(\text{sector}^2) \cdot n_C$ question: WHY rank^2 for 1-2 gap and N_c^2 for 2-3 gap? - Test Grace’s QCD-running reframe: does running $\times 0.70$ take $d(\nu)=64 \rightarrow$ observed 45 honestly? - Joint with Grace on running side - If closes: banks down mass ratios + sin θ_C via GST + θ_{12} linked pair

Lane L2 (parallel quick check): $A = 9/11$ substrate provenance - Grace's reading: $A = N_c^2/(2C_2 - 1)$, $-1 = T1444$ vacuum subtraction (forward across six observables) - Low-effort to verify; if banks $\rightarrow +1$ - Fresh lane, not fabrication-prone per Grace's filter

Lane L3 (background, multi-step joint): F436 propagator side for per-channel α deep gate (with Elie + Grace)

GRACE — Primary: QCD running mechanism for down-quark mass ladder

Lane G1 (primary, joint with Lyra): - QCD running side of $d(\nu)=64 \rightarrow 45$ IR bridge - Show $\times 0.70$ reduction is principled mechanism (not fit-tuning) - Falsifiable: if running doesn't take $64 \rightarrow 45$ cleanly, lane fails honestly

Lane G2 (parallel): sector-specificity mechanism investigation - Why three different mass-form structures (down=integer; lepton= π ; up=g)? - Boundary dictionary (down/lepton = bulk deposit; up = boundary determinant per Vol-16) - π -vs-integer split: substrate-mechanism explanation? Casey-named principle candidate territory.

Lane G3 (background): geometry side for per-channel α deep gate; Korányi-Wolf boundary stratification mining for un-extracted SM structure

ELIE — Primary: mass-side checker + cross-ratio survey

Lane E1 (primary): - Cross-check the (sector²)· n_C pattern across up-quarks and leptons (does the structure scale with running too, just sector-different?) - Toy 4521 confirmed down-quark; extend to up-quark and lepton ratios - Numerical/checker side for Lyra+Grace mass formula derivation; test machine confirms instantly when their derivation lands

Lane E2 (parallel quick check): - $A = 9/11$ numerical verification + null-model bracket (joint with Lyra) - $m_\tau/m_e = 49.71$ mechanism via $d(\nu)$ (re-routed from F437 kernel per Grace's category correction)

Lane E3 (background): per-channel α deep gate continues (Sakharov/cell-count side)

CAL — Post-landing review per Casey directive

- No in-loop gating; cold-read at landings only
- Casey will call when α tier-review is needed (F441 + Elie toy_4519 catch; recommendation “8 firm + 1 partial-forced (α value-banked Wyler, EM-identification gated)”)
- Per Cal #471 SYMMETRIC + Casey directive: seat outside live computation, later in time

KEEPER (me) — K-audit at landings + active form-space search

- Per K628 Keeper #26 sharpening: explicit substrate-primary form-space test BEFORE banking lean
- Tier-at-landing honestly; no “Cal-pending” framing
- Hold α at 9 baseline per Casey directive pending Cal review
- Active substrate-architectural investigation support (sector-specificity, dim_R cross-sector extension)

Linear algebra reminder (Casey standing order)

Every lane in matrix/operator form: - Vol-16 mass formula = $d(\nu) \times$ measurement-determinant matrix elements - QCD running = renormalization group flow on operator basis - $V_{CKM} = U_u^\dagger \cdot U_d$ literal matrix product - (sector²)· n_C pattern = eigenvalue ratios in diagonalized mass matrix - When stuck: write the matrix

Standing principles armed

- Casey directive: IGNORE CAL, DO NOT GATE, INVESTIGATE (Cal as later reviewer)
- Linearization standing order: every lane in matrix/operator form
- F417 (Casey-elevated): (B) mechanism-disambiguated vs (C) rich-vocab trap
- Cal #27 STANDING: fires hardest at peak elegance

- Cal #34 STANDING: don't promote empirical patterns
- Cal #35 STANDING: shared input → cross-check; distinct → forcing
- Casey #9 PRINCIPLED FLOOR: 4 pure scales open by scale-invariance of operator algebra
- Keeper #25: productive walk-back boundary
- Keeper #26: at peak motion, active substrate-primary form-space search
- Keeper #27: PRINCIPLED FLOOR not saturation
- [[feedback_pin_conventions_to_primary_sources]]: PDG values, not memory
- Grace's session memory: substrate-clean number-match is candidate, never bank, until mechanism delivers independently
- [[feedback_dont_manufacture_walls]] + [[feedback_no_fabricated_fatigue]]: investigate; stop only at honest content boundaries

What we don't pursue Wednesday

- More mixing-angle hunting without K-address grounding (high fabrication risk per Grace + Lyra explicit warnings; demonstrated this session)
- Re-litigation of θ_{12} 3/10 vs 5/16 (measurement-limited; resolves only when measurement tightens OR air-tight mechanism forces)
- Forcing $m_s/m_d = 20$ via $d(\nu)$ without QCD running (Lyra delivered honest negative; respect the finding)

Honest yield projection

Most likely: - Lane L1 partial close: +1 or +2 banks (m_b/m_s structural, or $A=9/11$) - Substantial structural progress on sector-specificity mechanism

Stretch: - Lane L1 closes cleanly via QCD running: +3 to +5 banks (down masses + $\sin \theta_C + \theta_{12}$) - $A=9/11$ closes parallel: +1 additional

Optimistic: - Per-channel α gate also lands: +1 + resolves α tier + closes F432 "why 5" - Net: 10 → 15+

Honest non-extension acceptable per directive: investigation matures; structural understanding deepens; count holds.

Pending Casey decisions

1. α tier-call (whenever ready; Cal review held per your directive)
2. θ_{13} formal banking (currently held at 9 baseline pending Cal α review; team consensus 10)
3. Direction-setting if joint mass-formula lane needs routing

— Keeper, Wednesday 2026-07-01 wake board. Down-quark mass formula via Vol-16 + QCD running is the highest-leverage Wednesday lane (joint Lyra + Grace). Substantial count motion (10 → 14+) reachable if mechanism closes; honest non-extension acceptable. Per Casey directive throughout.